DOCUMENT 11

Signal System Guide

Schema, ingestion API, hygiene flags, queries, and reliability weights.

NO GUESSWORK LIGHTING

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NGW

What is the Signal System?

Every shoot outcome a user records — "nailed it", "close", "failed", or "unknown" — is a signal. The Signal System classifies, tags, and filters these outcomes so that only the right signals reach each downstream consumer. The four inclusion flags are the key mechanism: they let you separate seeded bootstrap data from live user data without deleting any rows.

Database Schema

session_signals — full CREATE TABLE

```
CREATE TABLE session_signals (
  id             TEXT PRIMARY KEY,     -- UUID v4, client-generated
  session_id     TEXT NOT NULL,         -- links to the shoot session
  user_id        TEXT,                  -- nullable for anonymous
  pattern        TEXT NOT NULL,
    -- detected or target pattern name
  outcome        TEXT NOT NULL
    CHECK (outcome IN
      ('nailed_it', 'close', 'failed', 'unknown')),
  signal_source TEXT NOT NULL DEFAULT 'live'
    CHECK (signal_source IN
      ('live','seeded','internal','expert_review')),
  include_in_learning    BOOLEAN NOT NULL DEFAULT TRUE,
  include_in_metrics     BOOLEAN NOT NULL DEFAULT TRUE,
  include_in_conversion  BOOLEAN NOT NULL DEFAULT TRUE,
  include_in_cohorts     BOOLEAN NOT NULL DEFAULT TRUE,
  reliability_score      REAL CHECK (reliability_score BETWEEN 0 AND 1),
  region_scores TEXT,  -- JSON: {"face":0.91,"torso":0.78,...}
  degradation_flags TEXT, -- JSON: {"bw":false,"low_res":false,...}
  gear_tier   INTEGER CHECK (gear_tier BETWEEN 1 AND 5),
  environment TEXT,    -- studio | outdoor | hybrid
  created_at  DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

Signal Sources — Default Flag Values

source	in_learning	in_metrics	in_conversion	in_cohorts	Who Creates It
live	TRUE	TRUE	TRUE	TRUE	Real end users via app
seeded	FALSE	FALSE	FALSE	FALSE	Bootstrap / seed scripts
internal	FALSE	FALSE	FALSE	FALSE	Developer / QA testing
expert_review	FALSE	FALSE	FALSE	FALSE	Curator-verified outcomes

OVERRIDE

• The default flags can be overridden per-row at insert time or updated in bulk later. For example, expert_review signals can have include_in_learning = TRUE set manually to use them as high-quality learning data while still excluding them from public metrics.

Signal Ingestion API

POST /api/lab/signals accepts a batch of signal objects. Each object must include at minimum pattern, outcome, and signal_source. All other fields are optional.

POST /api/lab/signals — request body (JSON)

```json
{
  "signals": [
    {
      "id":            "sig_uuid_here",
      "session_id":    "sess_abc123",
      "user_id":       "usr_xyz",
      "pattern":       "loop",
      "outcome":       "nailed_it",
      "signal_source": "live",
      "gear_tier":     1,
      "environment":   "studio"
    }
  ]
}
```

POST /api/lab/signals — response (JSON)

```json
{
  "inserted": 1,
  "skipped":  0,
  "errors":   []
}
```

bash — ingest signals via curl

```bash
$ curl -X POST http://localhost:8000/api/lab/signals \
  -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
  -H "x-lab-email: you@org.com" \
  -H "Content-Type: application/json" \
  -d @signals_batch.json
```

Inclusion Flags — How Each Is Used

Flag	Consumer	Query Filter Applied

include_in_learning	Learning system (auto_candidate)	WHERE include_in_learning = TRUE
include_in_metrics	Dashboard analytics queries	WHERE include_in_metrics = TRUE
include_in_conversion	CVR calculations	WHERE include_in_conversion = TRUE
include_in_cohorts	User segmentation queries	WHERE include_in_cohorts = TRUE

SQL Query Reference

sql — learning-eligible signals by pattern (last 30 days)

```
SELECT
  pattern,
  outcome,
  COUNT(*)                             AS n,
  ROUND(AVG(reliability_score), 3)     AS avg_reliability,
  ROUND(AVG(gear_tier), 2)             AS avg_gear_tier
FROM session_signals
WHERE include_in_learning = TRUE
  AND signal_source       = 'live'
  AND created_at          >= DATE('now', '-30 days')
GROUP BY pattern, outcome
ORDER BY pattern, n DESC;
```

sql — metrics dashboard query (CVR per pattern)

```sql
SELECT
  pattern,
  COUNT(*) FILTER (WHERE outcome = 'nailed_it') AS nailed,
  COUNT(*) FILTER (WHERE outcome = 'close')     AS close,
  COUNT(*) FILTER (WHERE outcome = 'failed')    AS failed,
  COUNT(*)                                       AS total,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE outcome IN ('nailed_it','close'))
    / NULLIF(COUNT(*), 0), 1
  ) AS cvr_pct
FROM session_signals
WHERE include_in_metrics = TRUE
GROUP BY pattern
ORDER BY cvr_pct DESC;
```

sql — hygiene audit: check for mis-tagged seeded rows

```sql
-- Find seeded rows that were incorrectly flagged as live
SELECT id, session_id, created_at
FROM session_signals
WHERE signal_source = 'seeded'
  AND (include_in_metrics = TRUE
    OR include_in_learning = TRUE);

-- Fix: reset all seeded flags to FALSE
UPDATE session_signals
SET include_in_learning   = FALSE,
    include_in_metrics    = FALSE,
    include_in_conversion = FALSE,
    include_in_cohorts    = FALSE
WHERE signal_source = 'seeded';
```

Signal Reliability Weights

Each signal is scored across eight image regions. The raw score is multiplied by degradation factors that reduce weight when analysis conditions are poor.

Region	Weight	What It Detects
face	0.35	Primary lighting indicator — catch-lights, shadows, fill ratio
torso	0.15	Secondary — useful for full-length and 3/4 portraits
background	0.12	Fall-off curve, separation from subject
specular_surfaces	0.14	Modifier signature — size and hardness of specular highlights
shadow_regions	0.12	Shadow direction, density, and edge hardness
highlight_regions	0.12	Specular placement confirms key position

Degradation Factor	Multiplier	Trigger Condition
Black & white / monochrome	×0.75	Chroma saturation < 0.05
Heavy color grade	×0.80	Hue shift > 30° or LUT detected
No face detected	×0.70	Face region score == 0
Low resolution	×0.70	Image width < 512 px
Extreme contrast	×0.85	Histogram clipping > 5%
Environmental clutter	×0.90	Background region score < 0.3
Multiple shadow directions	×0.80	Solver finds > 1 shadow vector

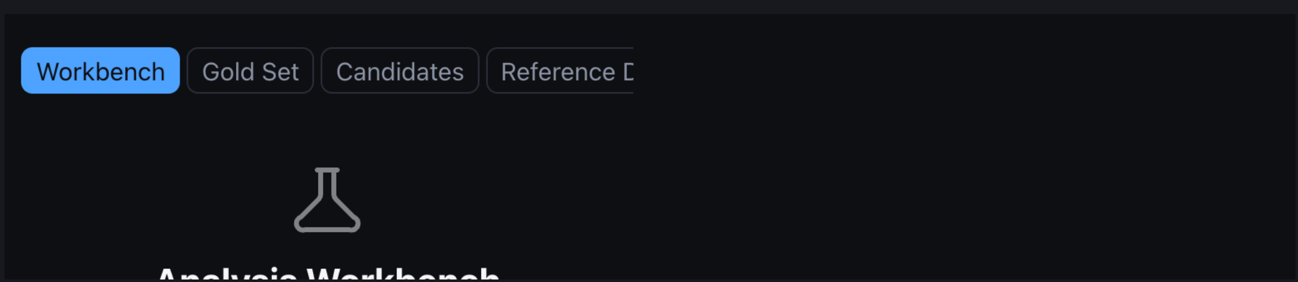
Signal Hygiene Summary API

GET /api/lab/signals/summary — response


```
{
  "live":              1482,
  "seeded":            500,
  "internal":          23,
  "expert_review":     8,
  "total":             2013,
  "learning_eligible": 1482,
  "metrics_eligible":  1482,
  "conversion_eligible": 1482,
  "cohorts_eligible":  1482,
  "flag_mismatches":   0
}
```


FLAG_MISMATCHES

- A non-zero flag_mismatches count means rows have signal_source="seeded" or "internal" but one or more include_* flags is TRUE. Run the hygiene audit SQL query above to identify and fix affected rows immediately.



Lab Workbench — Signal Hygiene summary card (live / seeded / internal / learning eligible)